Design of a Temperature-Compensated PLL Frequency Synthesiser

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Guide's Signature with Date

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Date: _____

Introduction & Problem Recap

- **Problem recap**
  - A PLL's ring-VCO drifts with temperature -- threshold voltage, mobility and capacitance all shift, so the same control voltage gives a different frequency hot vs. cold.
  - A compensation network cancels this by design, but sizing it needs a drift figure measured on the real topology, in the real process -- not assumed.
- **Objectives**
  - Design an integer-N charge-pump PLL (ref -> PFD -> charge pump -> loop filter -> 5-stage current-starved ring VCO -> /8 divider) in sky130, open-source, headless.
  - Measure uncompensated drift across all 5 process corners, then design and validate a temperature-compensation network with real statistical rigor (Monte Carlo + hypothesis testing).
- **Refinements after Review-I feedback**
  - Full 5-corner grid run (tt/ff/ss/sf/fs), not a 3-corner spot-check -- corrected the reported tempco spread from 3.4x to the real 12.5x (112-1399 ppm/C).
  - Cold-start acquisition risk (flagged as open in Review 1) fixed and verified: a true 0V cold start now locks in 1352 ns (retuned loop filter).
  - Added a second-order (curvature-matched) compensation network, novelty-checked against prior art before committing to the claim (see Literature Review).

Detailed Literature Review

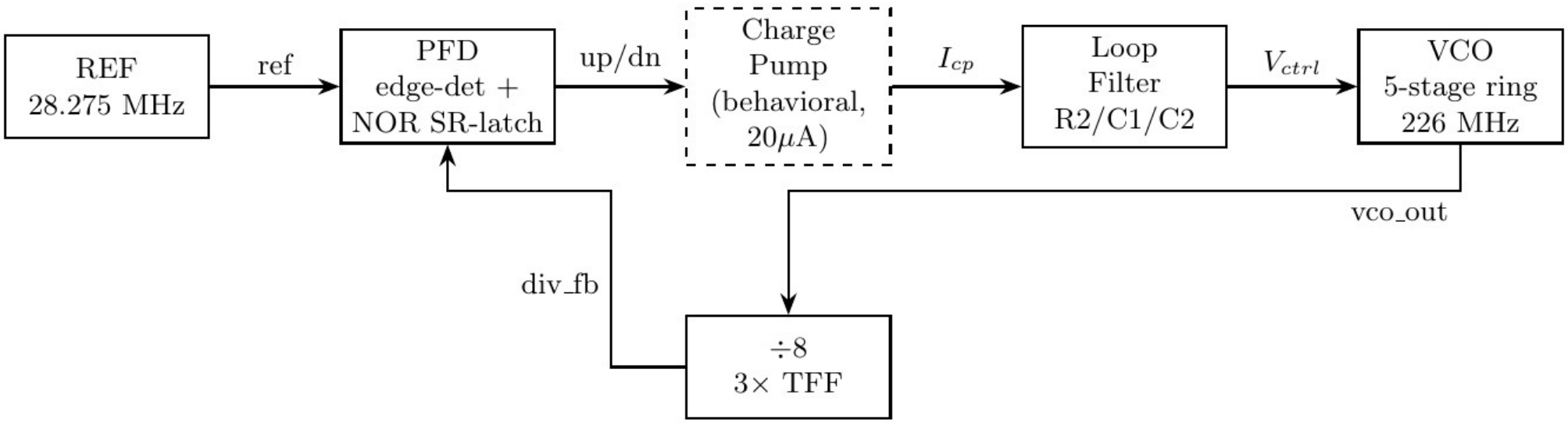
No.	Author(s) & Year	Title / Source	Method / Approach	Findings & Research Gap
1	G. Souliotis et al., 2022	Temperature compensated ring oscillator based VCO, AEU Int. J. Electron. Commun.	PTAT current compensation of a ring VCO inside a PLL; 180nm, 480MHz, fabricated	Closest topology benchmark: -1.5%/+0.4% over -40->120C, measured. First-order only; different node
2	C. Stancu et al., 2023	Temp. and power-supply compensated CMOS clock circuit based on ring osc., Electronics	Temp + supply compensated 50MHz ring-oscillator clock; simulation only	1.6% temperature error pre-layout. A clock circuit, never closed into a PLL
3	Z. Geng et al., 2018	Low power fast locking PLL freq. synthesizer with temp. compensation, EAME	Freq. presetting + dynamic loop BW + static/dynamic temp. comp.; 180nm, 2.1-2.6GHz	Lock <2us over 0-80C, -117dBc/Hz@1MHz, 9mW. Stability claim is the locked output, set by the crystal
4	Y. You et al., 2014	12GHz 67% tuning, 0.37ps RJrms PLL, LC-VCO temp. comp., 0.13um CMOS, IEEE RFIC	LC-VCO PLL for serial links; comp. scheme preserves tuning range; 5.6-13.4GHz	Stays locked -40->85C. Needs inductor models our process supports poorly; not portable to a ring VCO
5	S. Sinha, 2002	Design of an integrated CMOS PLL frequency synthesizer, IEEE MELECON	Dual-loop CP-PLL, 0.35um, 2.4-2.5GHz, 1MHz steps	-82dBc/Hz@10kHz. Architectural reference for our loop; no temperature compensation
6	E. M. V. Freitas et al., 2025	Using open source EDA tools in ASICs for HEP: a mixed comparison, JINST	Open-source vs commercial flows on IHP 130nm blocks incl. a current-starved ring VCO	Digital QoR quantified (+53% area, +118% power); analog VCO comparison qualitative, open-loop, no MC
7	M. Thiriloganathan et al., 2026	2.4GHz LC-VCO fractional-N PLL, open-source, 130nm BiCMOS, SMACD	Fully open-source flow (Xschem/ngspice/KLayout/OpenEMS), EM-modelled inductor	-100.8dBc/Hz@1MHz. Closes a loop open-source, but no temp. compensation, no PVT/Monte Carlo
8	R. N. S. Raphael et al., 2019	PLL-based frequency synthesizer for digital systems driving, BTSym'18 (Springer)	3rd-order CP-PLL for clock generation, integer-N (N=8-16), VCO input-cap-aware LF model	Settling <2us. Same loop class as ours; no temperature compensation

Detailed Literature Review (contd.)

No.	Author(s) & Year	Title / Source	Method / Approach	Findings & Research Gap
9	Y. Li, K. Du, J. Zhang, 2022	Design of ring oscillator with temperature compensation effect, Int. J. RF Microw. CAE	Positive-TC supply voltage on a 31-stage ring, 0.18um; 33kHz sleep timer	120 ppm/C (1.6%) over -40->80C. Single linear coefficient; kHz timer, not a PLL
10	H. H. Hammam et al., 2022	An ultra-low-power process-and-temperature compensated ring oscillator, IEEE ICEEE	3-stage current-starved ring, 130nm, process + temperature compensation	Up to 90% less variation vs. plain current-starved ring, 346nW. Standalone oscillator
11	S. Zhang et al., 2016	Temperature compensation technique for ring oscillators with tail current, Electron. Lett.	Tail current = constant source + PTAT current, first-order	Foundational first-order technique; no numbers reported in abstract; not closed-loop
12	M. Ortner, M. Punzenberger, 2017	Oscillators using quadratic temperature compensation, US Pat. 9,748,962 B2	Linear + quadratic temperature term for a VCO, -40->85C	Generic 2nd-order compensation is patented -- narrows our claim to ring-VCO + closed PLL + open PDK + stats
13	D. Patrinos, G. Souliotis, 2025	Low-voltage, low-power 2.5GHz ring oscillator with process & temp. comp., JLPEA	PTAT current + process-corner detection/tuning; 22nm, 0.8V; post-layout sim	Drift 23.9%->5.4% (tt) and 47.3%->7.2% (PVT). Corner-aware like our C(corner); not in a PLL
14	M. Sivasakthi, P. Radhika, 2024	PVT tolerant hybrid current starved ring VCO, bulk-driven keeper, 45nm, AEU	Two current-starved ring VCOs; Cadence GPDK45; corner + Monte Carlo	6.97GHz, -121.5dBc/Hz@1MHz. PVT tolerance of the VCO itself, no compensation network or loop
15	S. Madheswaran, R. Panneerselvam, 2024	7-stage MCML power-gated current starved VCO for PLL, IJECE	MOS current-mode logic delay cells with power gating, 45nm, simulation	3.06GHz, -125.9dBc/Hz. Same device class as ours; no closed-loop temperature characterisation

- Research gap (from all 15): no reported work combines a compensated ring VCO inside a closed loop, characterised across process corners with Monte Carlo rigor -- that gap is what this project fills.

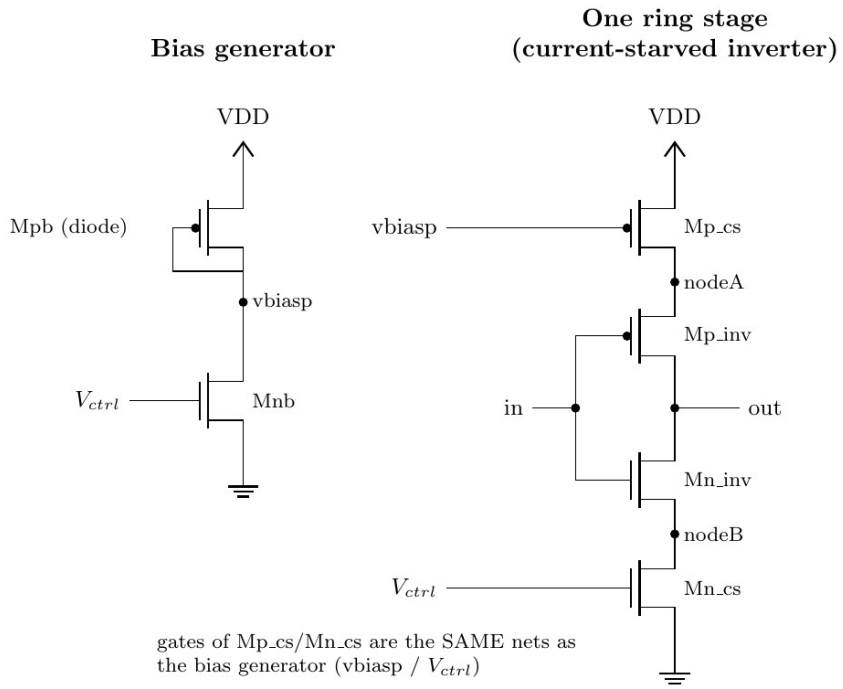
System Design & Architecture

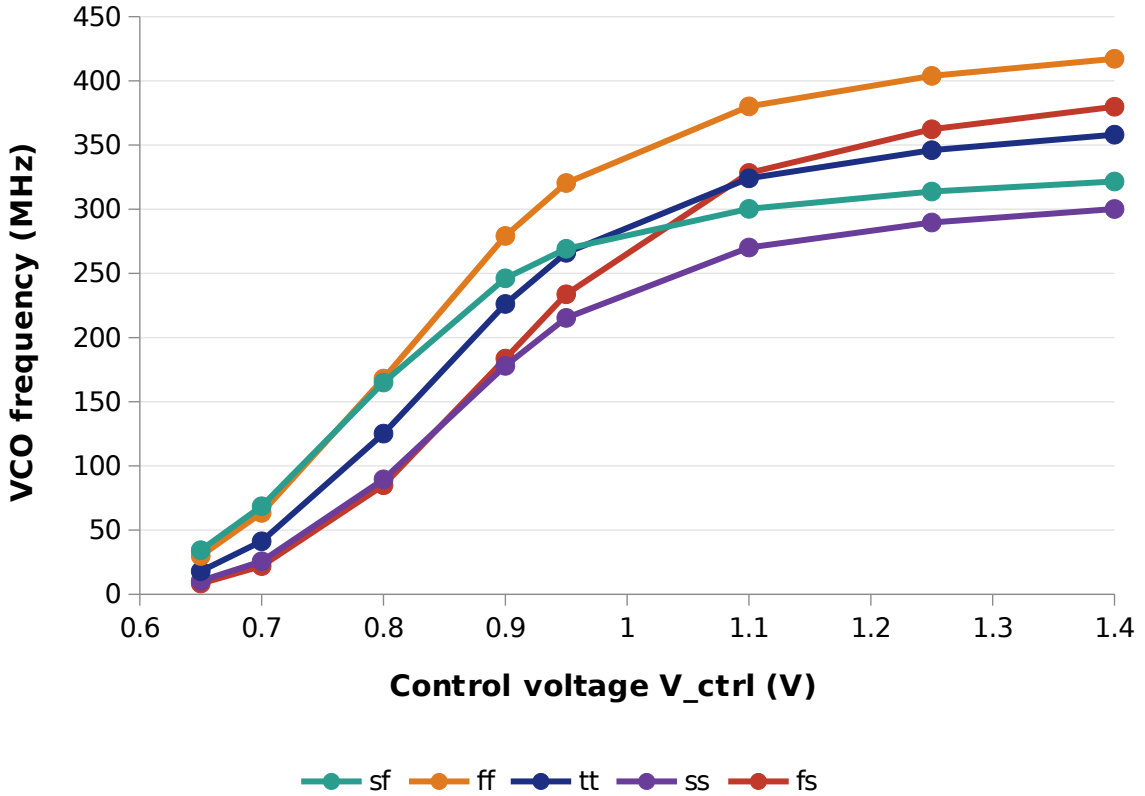


Closed-loop lock (tt, 27°C): **503 ns** (R2 = 3.6 kΩ; spec <2 μs)

- Integer-N charge-pump PLL: REF -> PFD -> charge pump -> passive loop filter -> 5-stage current-starved ring VCO -> /8 divider, with a compensation network on the VCO bias path.
- Scoping decisions (deliberate, documented): charge pump is an idealised 20uA behavioural source (transistor-level is Review 3, once layout needs real devices anyway).
- PFD uses edge-detectors + NOR SR-latches, not resettable DFFs -- a reliable async reset fights the latch's own bistable feedback loop.

VCO: Schematic & Tuning Curve





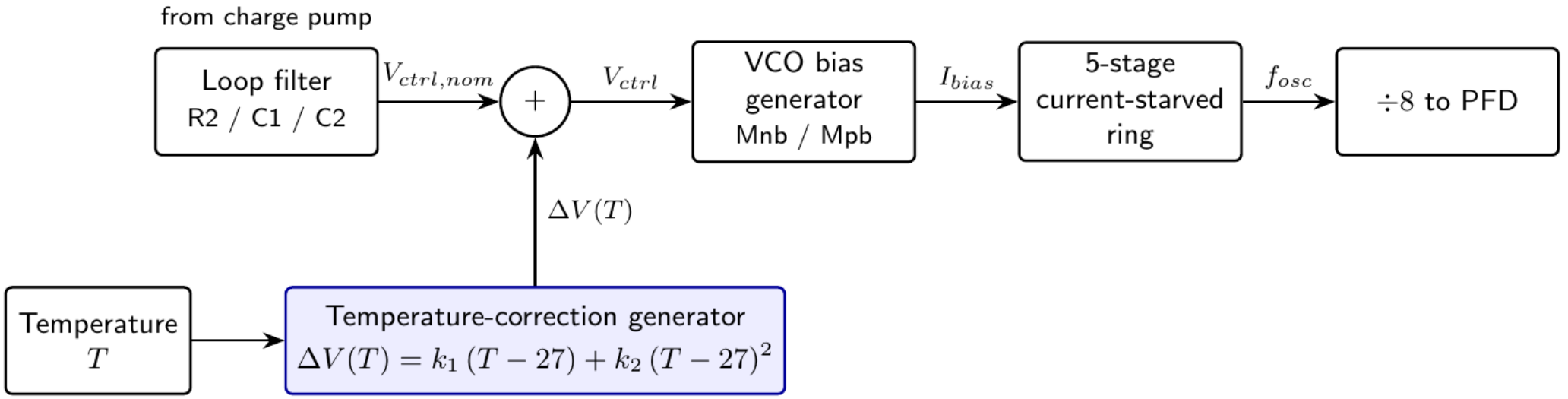
887 MHz/V
local K_VCO at 0.9 V (tt)

< 18 MHz
below 0.65 V (still oscillates)

5-stage current-starved ring: V_ctrl sets the starving current, so it sets the delay per stage.

Design point 0.9 V gives 226 MHz = 8 × 28.275 MHz reference.

Temperature-Compensation Architecture



1st-order: $k_2 = 0$ (linear, PTAT/CTAT-equivalent) **2nd-order:** $k_2 \neq 0$ (curvature-matched)
Coefficients fitted per process corner; an ideal behavioural source in Review 2.

ΔV(T)
correction added to V_ctrl before the VCO bias generator

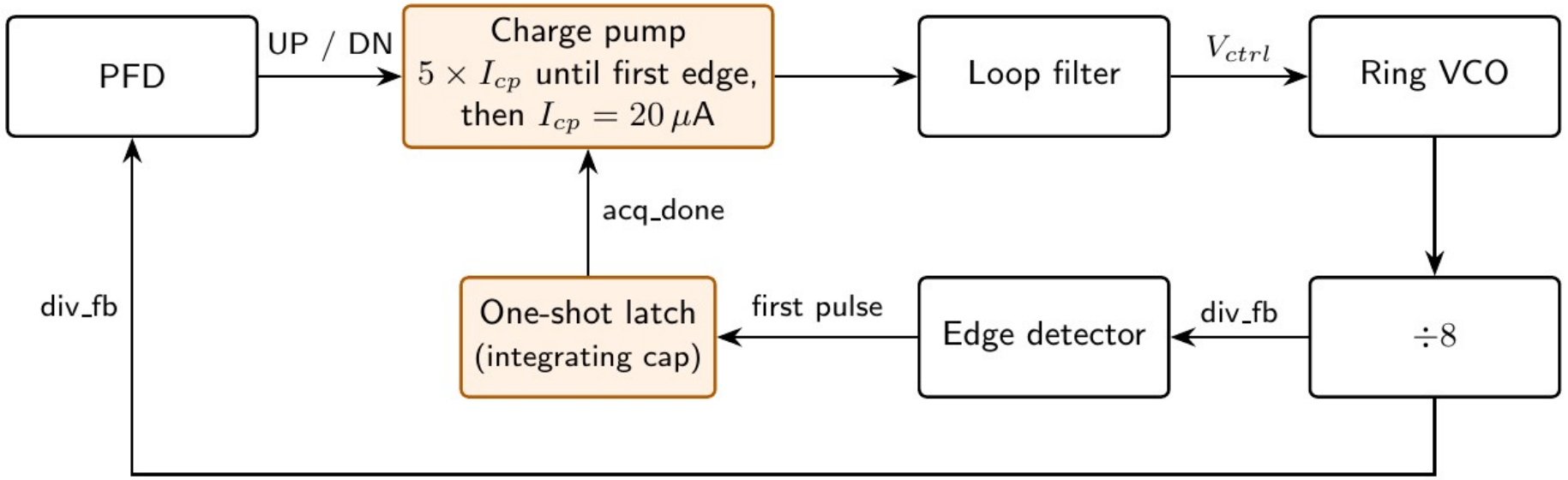
1st-order
linear k1·(T−27): PTAT/CTAT equivalent

2nd-order
adds k2·(T−27)²: matches the drift's curvature

per corner
k1, k2 fitted separately at tt/ff/ss/sf/fs

Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C (corner lock tests: coarse start).

Cold-Start Fix: Boosted Charge Pump



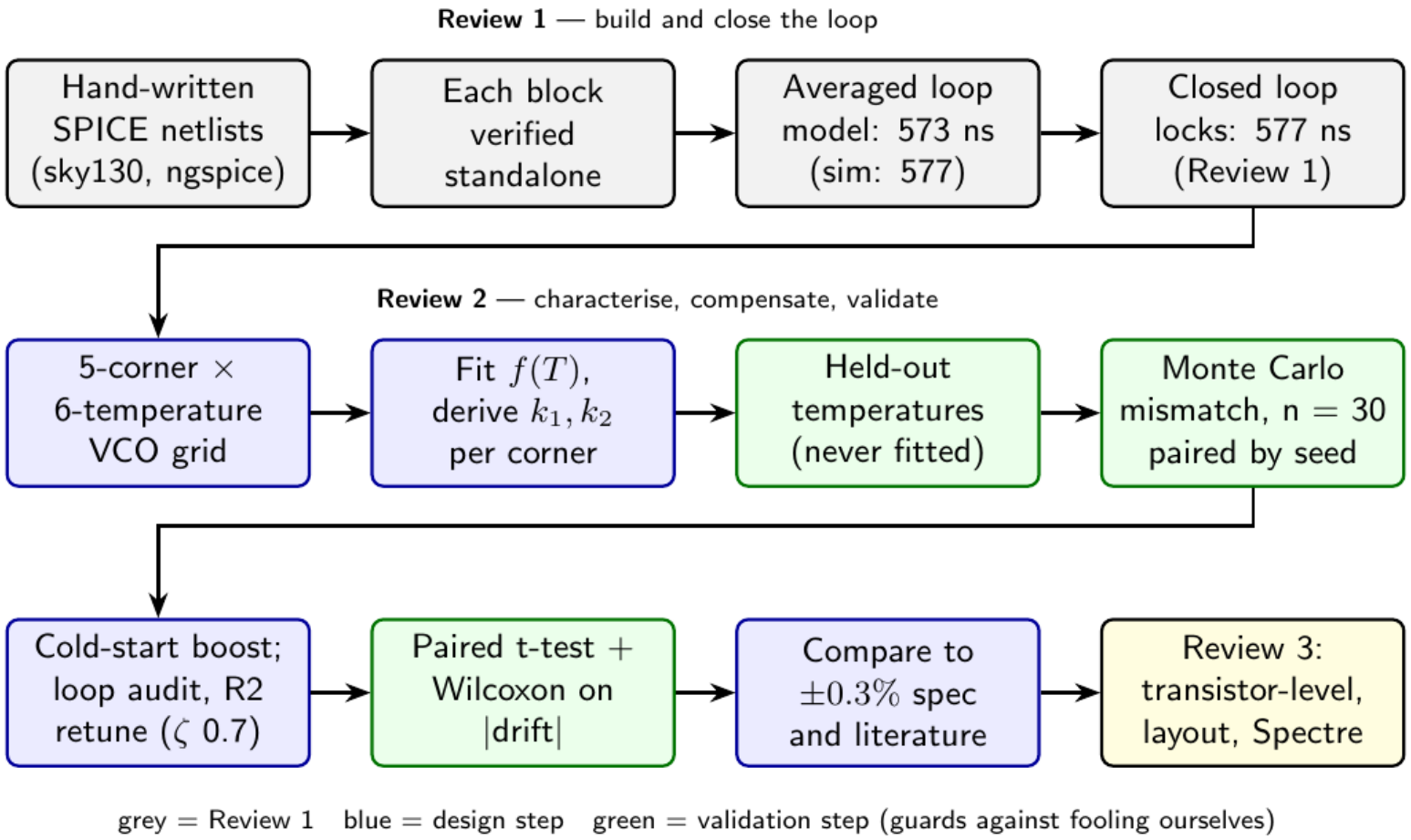
Problem (open since Review 1)

From a true 0 V start the VCO is in deep subthreshold (0.6 MHz at 0.5 V, 6.5 MHz at 0.6 V, vs 226 MHz target): the divided clock is effectively absent → PFD sees only REF → slow pump-up at 20 µA.

Fix

Charge pump runs at 5 × I_cp until the divider's first edge, then a one-shot latch drops it to the nominal 20 µA. Result: lock in 1352 ns from 0 V with the retuned loop filter (839 ns with the original), inside the 2 µs spec.

Methodology: Review 1 → Review 2 → Review 3



Implementation Details

- **Modules developed and integrated**
  - VCO (5-stage current-starved ring), PFD, charge pump, passive loop filter, /8 divider, digital primitive cell library (INV/TG/DFF/TFF) -- all hand-written SPICE, ngspice, headless.
  - Temperature-compensation network: 1st-order (PTAT/CTAT-equivalent) and empirically-refined 2nd-order (curvature-matched), as a parameterised behavioural correction voltage.
  - Acquisition-boosted charge pump (5x current until the divider's first edge) -- closes the cold-start dead-zone risk carried from Review 1.
- **Algorithms implemented**
  - Coefficient derivation: analytical 1st-order (from the measured f(T) polynomial fit and K_VCO), then empirical 2nd-order refit against the measured post-1st-order residual -- zero extra simulation for the refinement step.
  - Statistics: paired t-test, Wilcoxon signed-rank (normal approximation), OLS regression with slope confidence interval -- hand-implemented (no scipy in this environment), self-tested.
- **Code snippet -- the compensation correction (ngspice behavioural voltage source)**

```
Bcomp vctrl vctrl_nom V = 'k1*(T-27) + k2*(T-27)*(T-27)'
  ; k1 = -1.322e-4 V/C, k2 = 2.891e-7 V/C^2 (tt, empirically refined)
  ; verified via an isolated 4-line polarity test before use in the full loop
```

Results & Analysis (75%)

- VCO tuning: local K_VCO ~= 890 MHz/V at 0.9 V (tt, central difference +/-50 mV; an earlier whole-curve quadratic fit gave 669, too low); 18->358 MHz over 0.65->1.40 V. No hard floor: below 0.65 V the ring still oscillates, slowly (6.5 MHz at 0.6 V, 0.6 MHz at 0.5 V).
- Full 5-corner grid (all of tt/ff/ss/sf/fs, not a 3-corner spot-check) -- corrects the Review-1 finding: real tempco spread is 12.5x, not 3.4x. Worst corner (fs) wasn't checked before.

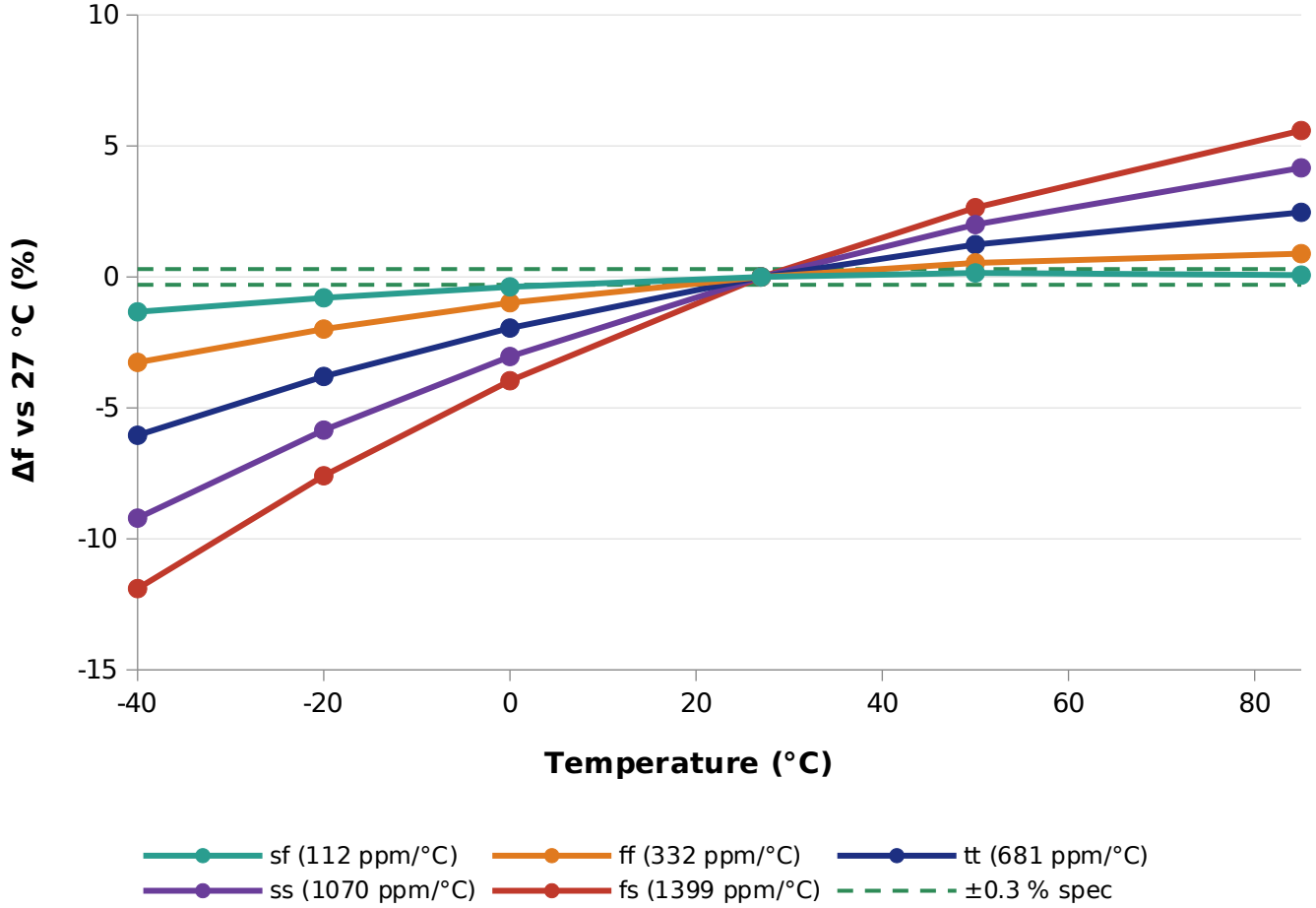
Corner	NMOS/PMOS	K_VCO (MHz/V)	Uncompensated tempco
sf	fast / slow	582	+112 ppm/C (flattest)
ff	fast / fast	936	+332 ppm/C
tt	typical / typical	887	+681 ppm/C
ss	slow / slow	815	+1070 ppm/C
fs	slow / fast	1018	+1399 ppm/C (steepest)

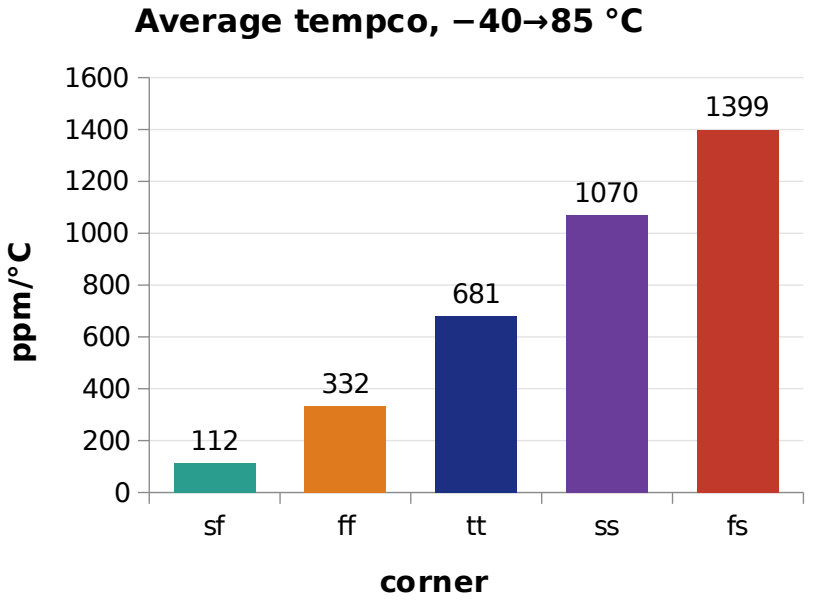
12.5x
spread, not the 3.4x previously reported

887
MHz/V K_VCO
(tt, 0.9 V, local slope)

- Tempco ranks with NMOS drive at 0.9 V (sf 45 uA ... fs 28 uA): less overdrive => the threshold drop with temperature dominates => stronger positive tempco. Sign flips at V_ctrl = 1.4 V (-816 ppm/C). sf/fs bracket the range because sky130 skew corners shift NMOS Vth +/- 27 mV (ss/ff +/-15 mV).

Uncompensated Drift: All 5 Corners



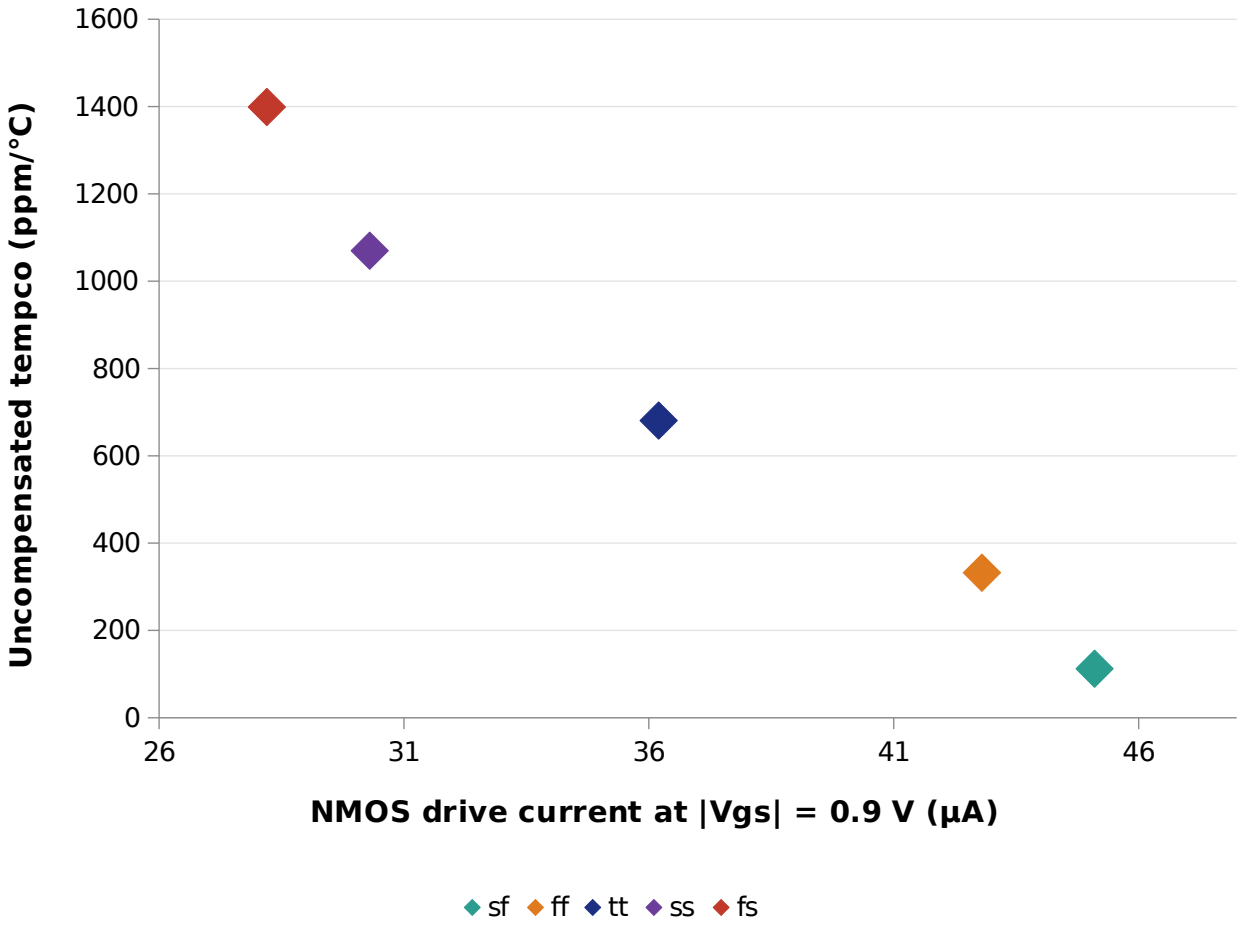


12.5×
tempco spread sf→fs

8.5 %
tt span, −40→85 °C

Every corner is far outside ±0.3 % uncompensated. Tempco ranks with NMOS drive: see next slide.

Why the Corners Order This Way



Reading the chart

V_ctrl drives the NMOS starving gate directly and sets the PMOS bias through a mirror, so NMOS overdrive sets the ring current.

Less overdrive → the threshold-voltage drop with temperature is a bigger share of it and beats the mobility drop → stronger positive tempco.

Corner labels in this PDK

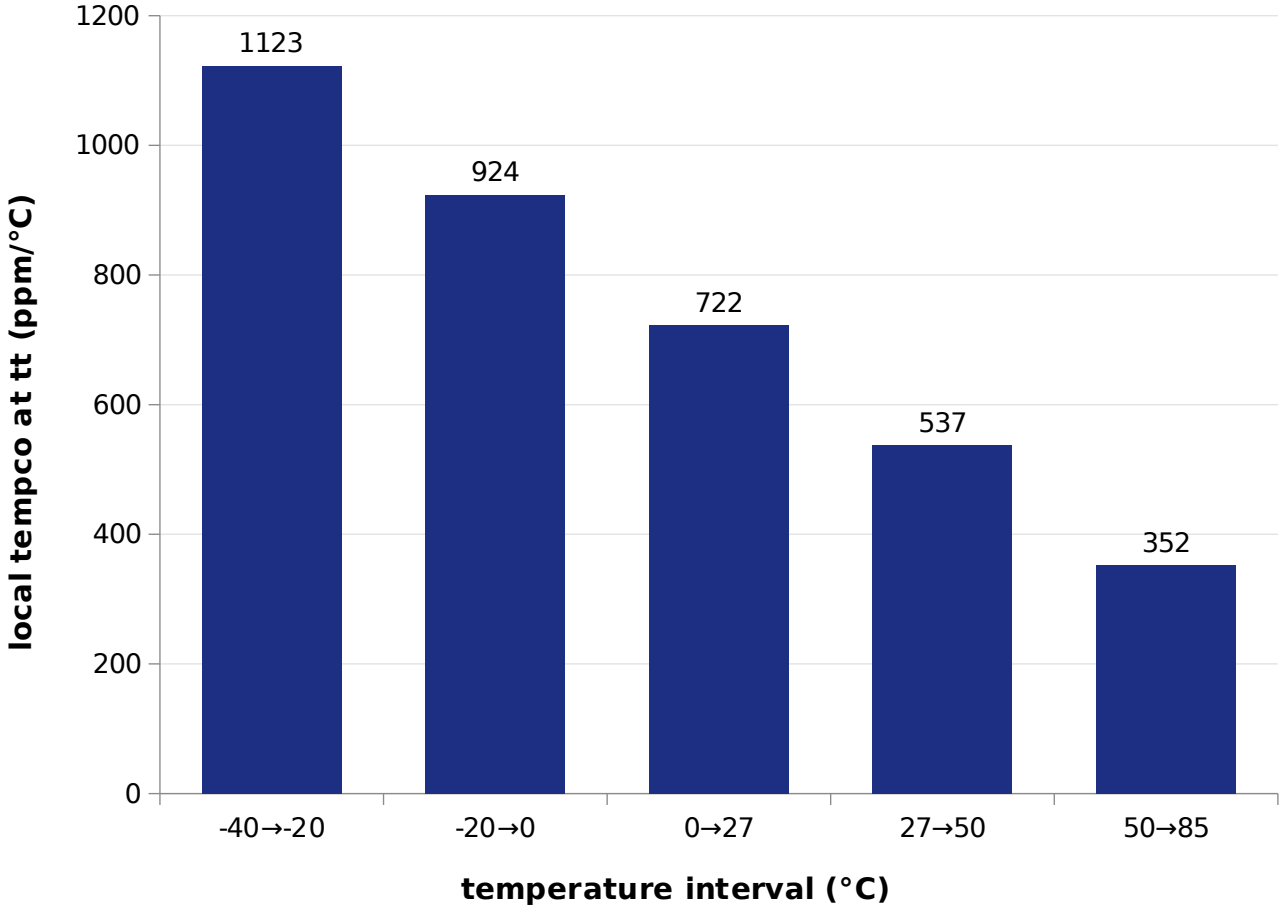
sf = fast-NMOS / slow-PMOS, fs = slow-NMOS / fast-PMOS. The skew corners shift NMOS Vth ±27 mV (ss/ff only ±15 mV), so they bracket the range.

−816 ppm/°C
tt tempco at V_ctrl = 1.4 V (sign flips)

+681 ppm/°C
tt tempco at V_ctrl = 0.9 V

Sign flip = a zero-tempco bias point exists between 0.9 and 1.4 V (textbook behaviour).

Why Second Order? The Drift Is Curved



Reading the chart

If the drift were a straight line, every bar would be the same height (the 681 ppm/°C average).

Instead the slope falls 3.2× from cold (1123) to hot (352) — the drift is curved.

Consequence

A linear (1st-order) correction can only match the average slope: it leaves a bowed residual.

A quadratic term $k2 \cdot (T-27)^2$ matches the curvature → 2nd-order compensation.

Results & Analysis (contd.)

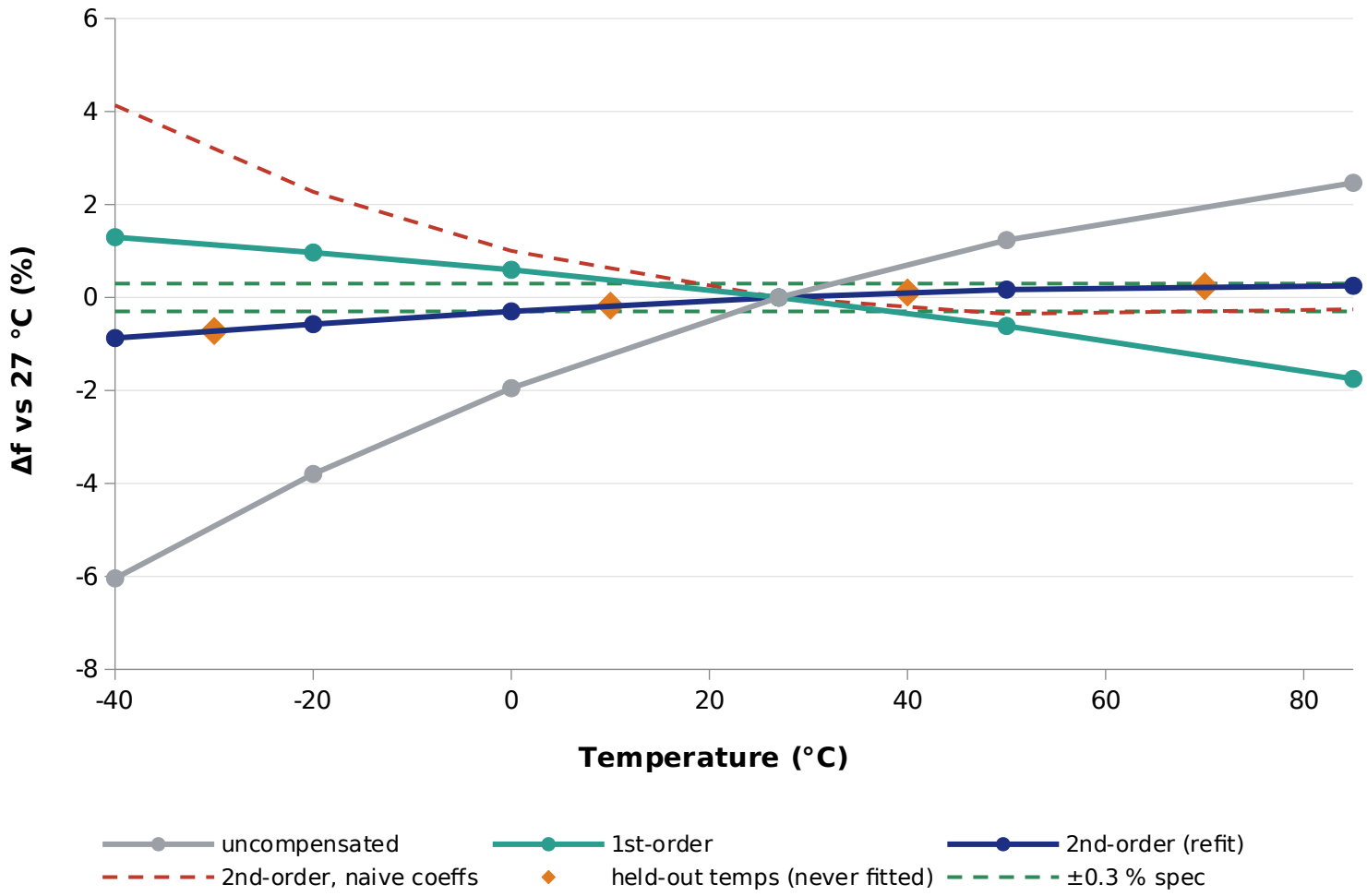
- Compensation refit independently at every corner; every corner improves at both stages. Checked at 4 held-out temperatures never used in fitting (-30/10/40/70C): all stay inside the fitted envelope.

Corner	Baseline span	1st-order span	2nd-order span	2nd-order max |dev|	Meets +/-0.3%?
ff	4.15%	1.06%	0.10%	0.07%	Yes
sf	1.48%	0.42%	0.24%	0.15%	Yes
tt	8.51%	3.05%	1.12%	0.87%	No
ss	13.37%	6.37%	3.32%	2.46%	No
fs	17.49%	9.13%	5.31%	3.83%	No

- **Our +/-0.3% target is met at ff and sf only -- not at the typical corner. Closing the gap needs a transistor-level network whose PTAT slope tracks process, and/or a per-corner trim (Review 3).**
- Naive analytical 2nd-order coefficients made tt worse (4.49% vs 3.05%): they assume constant K_VCO (false) and used a K_VCO ~25% too low (669 vs 887 MHz/V). Refitting against the measured 1st-order residual fixed it.

Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C.

Compensation at the Typical (tt) Corner



8.51 → 1.12 %
span: baseline → 2nd-order

7.57×
compensation ratio C at tt

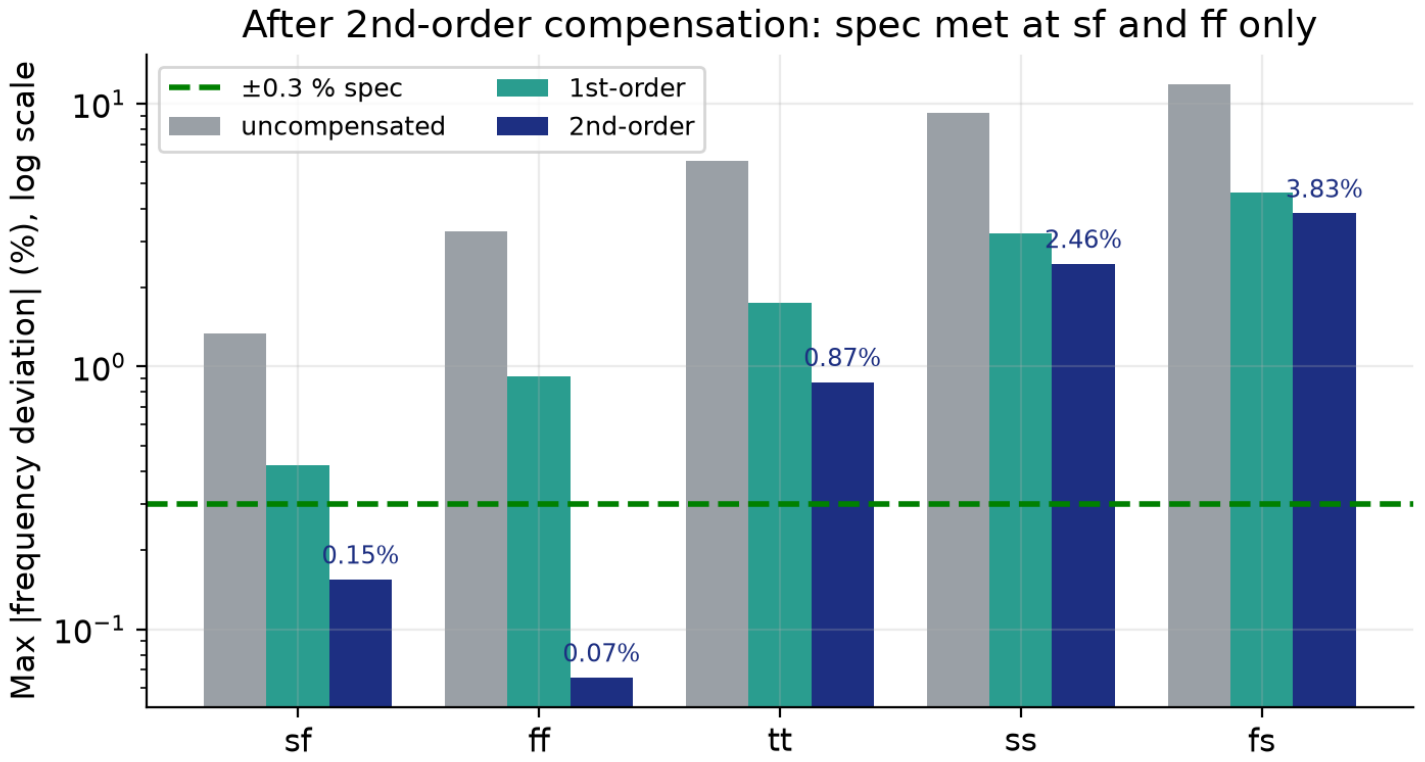
±0.87 %
max |dev|: spec ±0.3 % NOT met

Naive analytical k2 over-corrects the cold end (+4.1 % at −40 °C); refitting on the 1st-order residual fixes it.

Orange diamonds: temperatures never used in fitting, all inside the envelope.

Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C (corner lock tests: coarse start).

Residual Drift per Corner vs ±0.3 % Spec



ff: 0.07 %
meets spec

sf: 0.15 %
meets spec

tt: 0.87 %
misses spec

ss: 2.46 %
misses spec

fs: 3.83 %
misses spec

Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C (corner lock tests: coarse start).

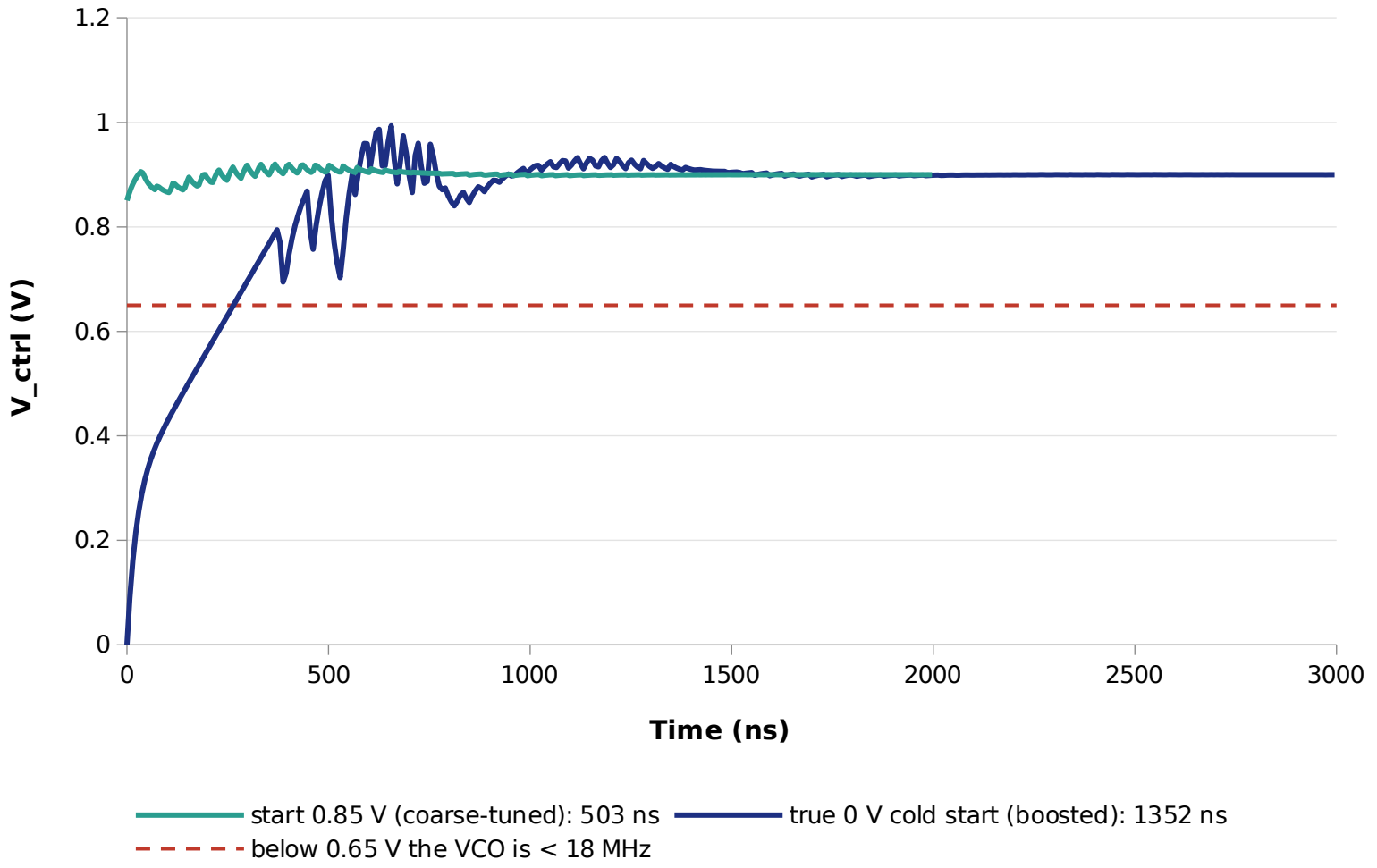
Results & Analysis (contd.) -- Closed-Loop Lock

Quantity	Coarse-tuned start (0.85V), R2 = 3.6k	TRUE cold start (0V, boosted), R2 = 3.6k
Lock time	503 ns (was 577 ns)	1352 ns (was 839 ns)
Settled V_ctrl	0.9000 V (was 0.902)	0.9000 V
Settled f_VCO	226.18 MHz (was 227.96)	225.98 MHz (was 226.42)
Spec / prediction	< 2us spec	< 2us spec

- Cold-start acquisition dead zone (open risk since Review 1) -- fixed and verified. Built a 5x-boosted-current charge pump variant that reverts to nominal current once the divider produces its first edge (tracked via a B-source-charged integrating capacitor, reusing the same current-injection idiom already proven correct -- not a new digital latch and its metastable-initial-state risk).
- Simulated from a TRUE 0V cold start (not the coarse-tuned 0.85V assumption): locks in 1352 ns with the retuned loop filter (839 ns with the original R2 = 1.7 kOhm), inside the 2us spec.
- Two real bugs found and fixed while closing the original loop (Review 1): an inverted SPICE B-source node-order convention (caught with a 5-line isolated test before touching the full loop), and an inconsistent loop-filter capacitor initial condition (a 50ns charge-pump hold gate was added to test a wrong hypothesis first -- disproving it is what pointed at the real cause).

Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C.

Closed-Loop Lock Transient (Retuned Loop)



1352 ns
lock from 0 V cold start (was 839 ns)

503 ns
lock from 0.85 V (was 577 ns)

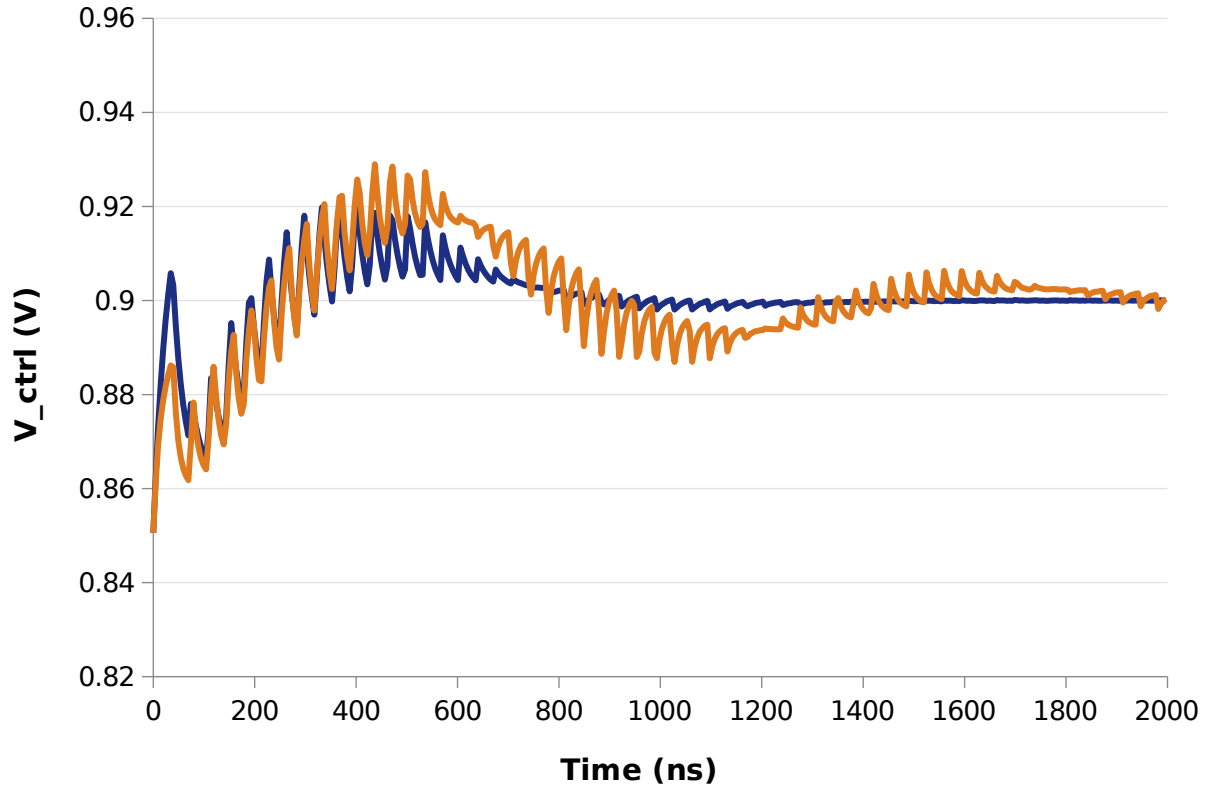
< 2 µs
spec: met at both starts

Lock = V_ctrl within ±2 % of 0.9 V and staying there.

R2 = 3.6 kΩ. The retune makes the cold start slower (+61 %) but well damped.

Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C (corner lock tests: coarse start).

Loop Dynamics: Found, Fixed, Verified



Measured local K_VCO = 887 MHz/V (Review 1 used 669).

ζ 0.33 → 0.70
damping (Review 1 claimed 0.72)

≈ 30° → 65°+
phase margin (100·ζ rule)

8.6 → 0.35 mV
V_ctrl ripple at lock

0.78 → 0.008 %
residual frequency error at 2 µs

Found

Review 1 dropped the 1/(2π) of the charge-pump gain, so ζ was really 0.33, not 0.72.

Fixed

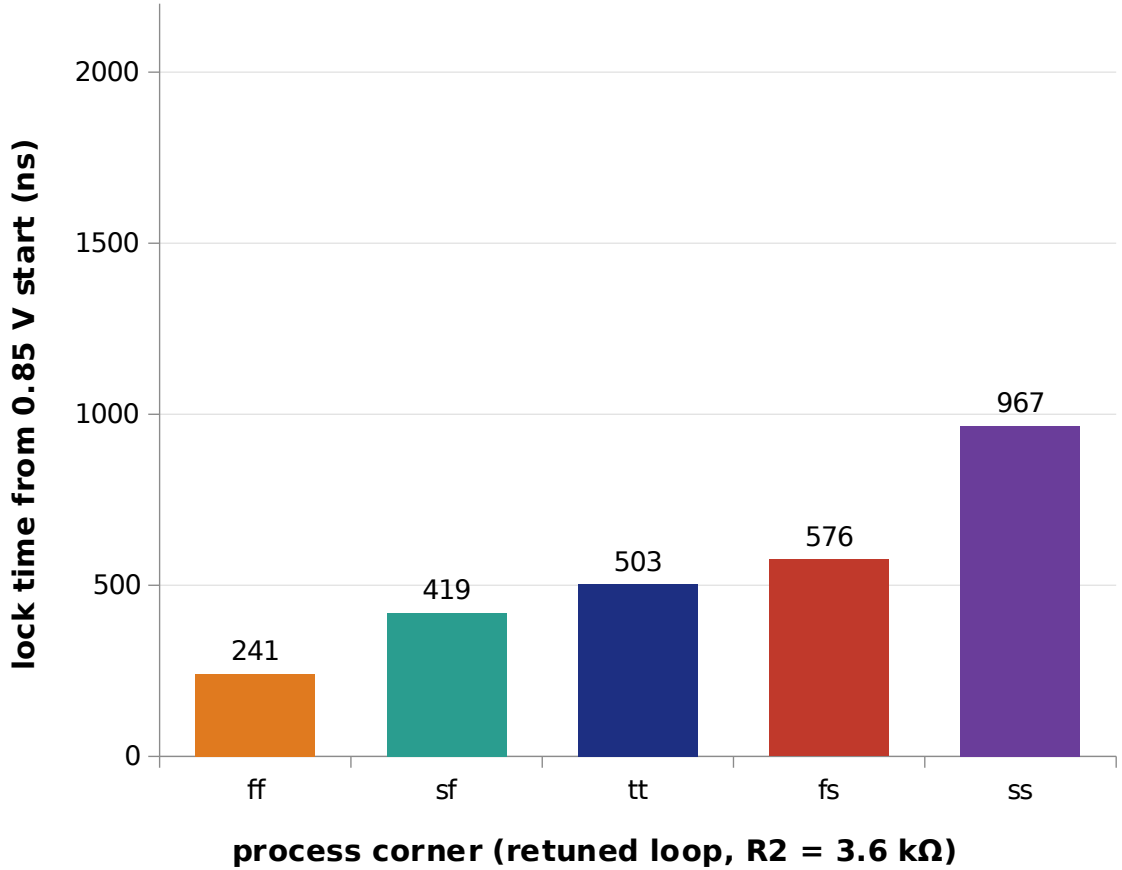
R2 1.7 → 3.6 kΩ (C1, C2, I_cp unchanged), verified at transistor level.

Cost

0 V cold start 839 → 1352 ns; both inside the 2 µs spec.

Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C (corner lock tests: coarse start).

Lock Across All Five Corners



Retuned loop: all five corners lock inside the 2 µs spec

Settled V_ctrl differs by corner (ss needs 0.968 V, the weakest NMOS); final f_VCO within 0.1 % of 226.2 MHz at ff, sf, tt, fs.

corner	retuned lock	final f_VCO	original loop
ff	241 ns	226.2 MHz	not settled at 2 µs (225.2 MHz)
sf	419 ns	226.4 MHz	not settled at 2 µs (229.1 MHz)
tt	503 ns	226.2 MHz	577 ns
fs	576 ns	226.2 MHz	not settled at 2 µs (228.9 MHz)
ss	967 ns	settled (5 µs run)	1737 ns (5 µs run)

The original loop had not finished settling at ff/sf/fs/ss within the 2 µs run, so no lock time is quoted for it there. Corner runs use the coarse start; the 0 V cold start was run at tt only.

Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C (corner lock tests: coarse start).

Results (contd.) -- Monte Carlo & Benchmark

Arm (tt, mismatch on, paired by seed)	n	Mean |drift|, 85C vs -40C	95% CI
Baseline (uncompensated)	12	18.93 MHz (8.4%)	[16.20, 21.65]
1st-order	30	5.85 MHz (2.6%)	[4.62, 7.09]
2nd-order (curvature-matched)	30	3.37 MHz (1.5%)	[2.33, 4.41]

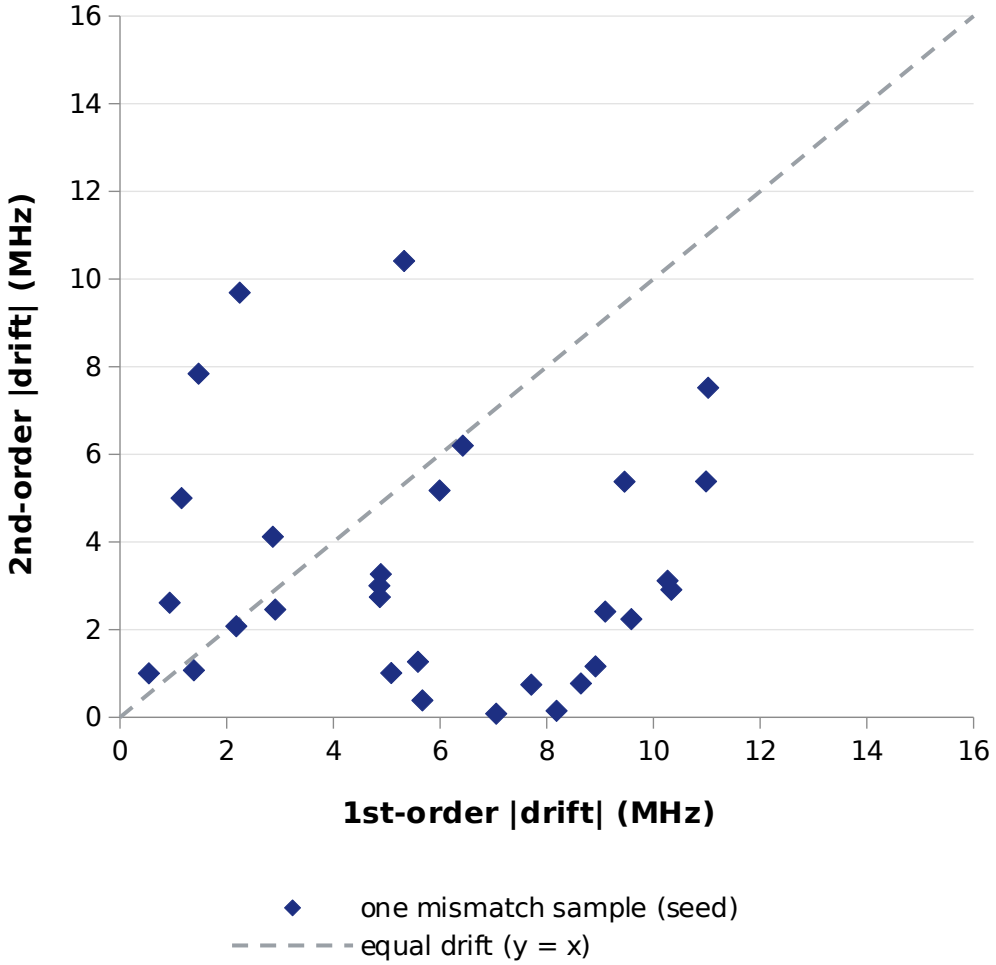
- **Both stages significantly reduce drift (12/12 seeds). 2nd-order beats 1st-order: -2.48 MHz, 95% CI [0.81, 4.15], paired t-test and Wilcoxon agree, 23/30 seeds.**
- Correction on record: our first pass tested signed drift and overstated this at n=12 (1st-order over-corrects, so the sign flipped). On |drift| it was not significant at n=12 -- n=30 is what earns the claim.

Design	Process	Residual drift	Lock time
This work, 2nd-order (tt)	sky130, simulation	+/-0.87% max dev., -40 to 85C	1352 ns (0V cold start)
Souliotis et al. 2022 [1]	180nm, measured silicon	-1.5% / +0.4%, -40 to 120C	not reported
Stancu et al. 2023 [2]	simulation, 50 MHz clock	1.6% temperature error	n/a
Geng et al. 2018 [3]	180nm, measured silicon	not reported as drift	< 2 us

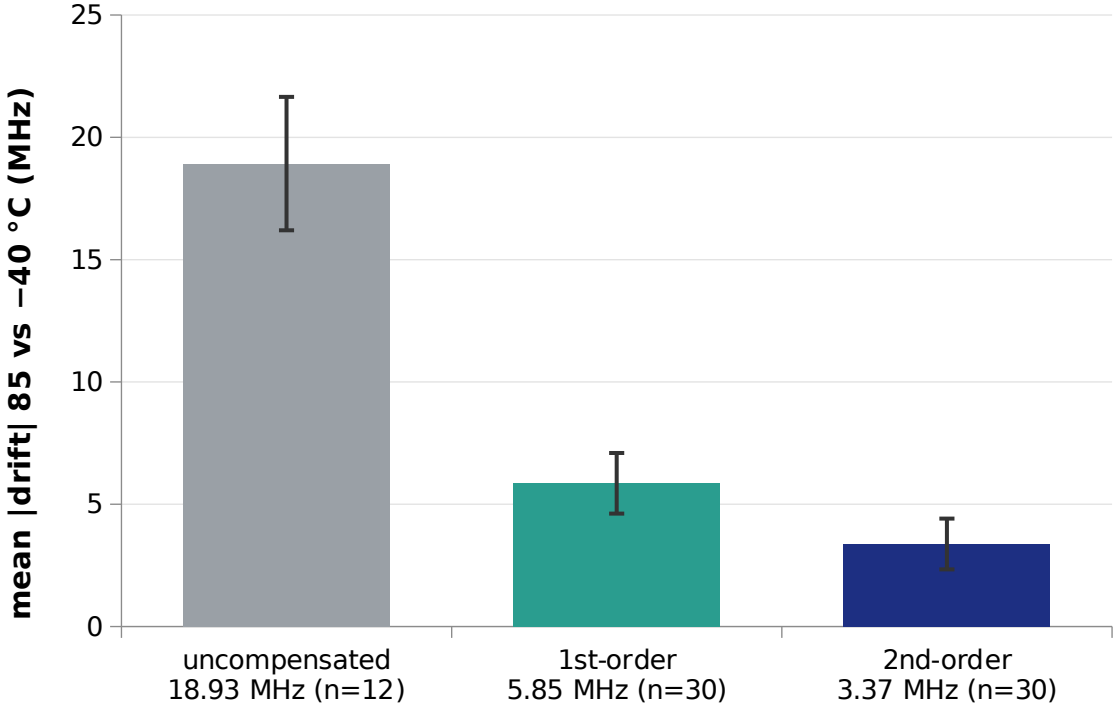
- Comparable to published work, but ours is simulation of an ideal correction network vs. their measured silicon -- not a like-for-like claim until the transistor-level network exists.

Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C.

Monte Carlo: Paired Comparison (tt)



Points below the dashed line: 2nd-order beat 1st-order on that same chip.



23 / 30
seeds where 2nd < 1st

−2.48 MHz
95 % CI [0.81, 4.15]

p < 0.05
paired t + Wilcoxon

Review 2 at a Glance

12.5×
tempco spread across 5 corners (112→1399 ppm/°C)

887 MHz/V
local K_VCO at 0.9 V, tt

1.12 %
tt span after 2nd-order (was 8.51 %)

2 / 5
corners meeting ±0.3 % (ff, sf)

3.37 MHz
Monte Carlo mean |drift| after 2nd-order (was 18.93)

23 / 30
seeds where 2nd-order beats 1st

1352 ns
lock from a true 0 V cold start, retuned loop (< 2 µs)

15
IEEE references; 1 patent narrowed our novelty claim

Scope: compensation is an ideal behavioural correction voltage (transistor-level in Review 3); closed-loop results are tt / 27 °C (corner lock tests: coarse start).

Challenges & Remaining Work

- **Challenges faced and solutions**
  - SPICE B-source polarity and an inconsistent loop-filter initial condition broke the first closed-loop runs -- each isolated with a minimal test (and one wrong hypothesis disproved) before touching the full loop.
  - Naive analytical 2nd-order coefficients over-corrected the cold end -- fixed by refitting against the measured 1st-order residual; the refit was then checked at 4 held-out temperatures it never saw.
  - Caught our own statistics error: the first Monte Carlo pass compared SIGNED drift, which made 1st- vs 2nd-order look significant only because the sign flipped. Re-tested on |drift| and re-run at n=30.
  - Novelty check found a 2017 patent on generic quadratic compensation -- narrowed our claim to ring-VCO + closed-loop PLL + open PDK + statistically validated quantification.
- **Remaining work (25%)**
  - Close the spec gap: +/-0.3% is met at ff and sf only (tt 0.87%, ss 2.46%, fs 3.83%) -- a transistor-level compensation network whose PTAT slope tracks process, plus a per-corner trim.
  - Re-verify the retuned loop (zeta ~0.7, R2 = 3.6 kOhm) with a transistor-level charge pump; layout in Magic, DRC, LVS (netgen), post-layout re-simulation.
  - Cadence/Virtuoso cross-validation arm and the actual H1 regression (eps_res vs C) -- the other half of the research hypothesis, pending lab access.
- **Timeline to Review-III (28.10.2026)**
  - Weeks 1-2: transistor-level CP + compensation network. Week 3: layout / DRC / LVS. Week 4: Virtuoso arm, H1 regression, final report.

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Thank You

Project-I (2026) · SENSE · VIT Chennai · github.com/pocopepe/visualizer